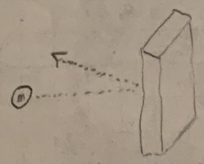
Ball @ door Lab | Oliver Ohrt 3/9/22 | period 2

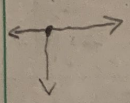
What we know

Given 3 different balls with the same mass (m) and thrown at the same velocity (v): Happy ball, which hits the door and bounces off Sad Ball which hits the door and stops, and clay ball which hits the door and sticks. Which ball will cause the door to close the fastest?

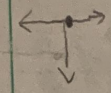


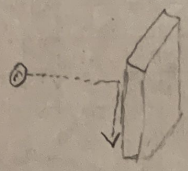
[happy ball]

Before:



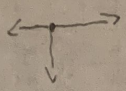
After:





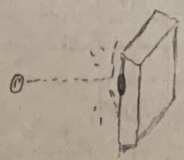
[Sad ball]

Before:



After:





[Clay ball]

Before:

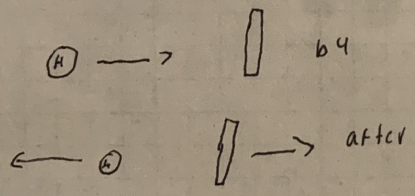
After:

splats on door?



because the happy ball bounces off it must have a greater impulse

$$\vec{P}_{sys} = \vec{P}_1 + \vec{P}_2 + \vec{P}_3 \ldots$$



* because the ball continues moving means greater change in P

Oliver Ohrt

Mr. Brookshire

Physics

03 March 2022

Discussion

From the diagrams I made, and the discussions we had in class, I can infer that the Happy Ball hitting the door is an elastic collision. This means that there is no loss of kinetic energy, like Newton's Cradle. You could also think of it as two different reactions. One where the ball hits the door, and one where the ball bounces off of the door. Unlike the clay or sad ball, which only have one reaction. The initial Force, Acceleration, and Momentum are all the same for the happy, sad, and clay, ball. However after impact with the door the sad and clay ball lose all momentum. But the happy ball bounces off of the door causing the door to have an **equal and opposite reaction, keeping** the balls' momentum and causing the door to close faster.

Ultimately, the rubber "Happy Ball" will close the door the fastest because it induces an elastic collision which conserves kinetic energy. The impulse which is the change in momentum is greater for the happy ball as the ball conserves its velocity after hitting the door. And a greater impulse means a greater velocity from the reactant side, which in this case is the door.